

✓ Binary to Decimal

%%writefile binarydecimal.cpp

```
#include<iostream>
#include<math.h>
using namespace std;
int main(){
    int n;
    cin>>n;
    int ans=0;
    int i=0;
    while(n!=0){
        int digit=n%10;
        if (digit==1){
            ans+=pow(2,i);
        }
        n/=10;
        i++;
    }
    cout<<"Answer is: "<<ans<<endl;
}
```

 Writing binarydecimal.cpp

```
!g++ binarydecimal.cpp -o binarydecimal
!./binarydecimal
```

 111
Answer is: 7

✓ Decimal to Binary

%%writefile decimalbinary.cpp

```
#include<iostream>
#include<math.h>
using namespace std;
int main(){
    int n;
    cin>>n;
    int ans=0;
    int i=0;
    while(n!=0){
        int bit=n&1;
        ans=(bit*pow(10,i))+ans;
        n=n>>1;
        i++;
    }
    cout<<"Answer is: "<<ans<<endl;
}
```

 Writing decimalbinary.cpp

```
!g++ decimalbinary.cpp -o decimalbinary
!./decimalbinary
```

 16
Answer is: 10000